

XRPL Validator — Daily Engineering Report • 2026-09-05

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 25 commits (xrpl-validator 25)
- 4 engine findings fixed (F181–F184)
- Amendment **fixCleanup3_3_0** held majority (since 2026-08-28 11:15 UTC); activation due ~2026-09-11 11:15 UTC

Engine findings fixed

- **fix(xrpl-ledger): F181 — a holder-less sender ISSUES to the issuer within its limit** (30d4e7a , xrpl-validator) `apply_iou_direct` sized the sender's leg by `available()` — the sender's HOLDINGS of the issuer's IOU — so a sender holding none was `tecPATH_DRY`. rippled's sender hop is `DirectStepI::maxPaymentFlow` (`DirectStep.cpp:476-490`): when `srcOwed <= 0` the hop ISSUES the sender's own IOU up to `creditLimit2(dst, src) + srcOwed`, the issuer's trust limit toward the sender less what the sender already owes it.
- **fix(xrpl-ledger): F182 — CredentialCreate needs the reserve for one more object** (944aeb1 , xrpl-validator) The native `CredentialCreate` had no reserve check and no subject check: it wrote the credential, both owner-directory pages and the `OwnerCount` bump for any issuer. rippled (`CredentialCreate.cpp`) answers `tecNO_TARGET` when the subject account does not exist (preclaim, before the duplicate test) and, in `doApply` before any write, `checkReserve(ctx, sleIssuer, preFeeBalance_, {.ownerCountDelta = 1})` — the balance BEFORE ...
- **fix(xrpl-ledger): F183 — a strand's spend is the walk's own figure, not the line's quantised delta** (5449dfa , xrpl-validator) `strand_pass` measured what a strand spent by differencing the sender's line before and after, trusting the walk's own figure only when the two agreed to 1e-9 relative. A trust line holds 16 significant digits on its `MAGNITUDE`, so the delta is the true spend quantised to the balance's ulp — for a small spend from a large line that quantum dwarfs the relative test and the exact figure was refused. rippled never reads ...
- **fix(xrpl-ledger): F184 — an autobridged leg crosses the taker's own offer** (199e7d4 , xrpl-validator) The self-cross rule (`BookOfferCrossingStep::limitSelfCrossQuality`, `BookStep.cpp:402-445`) removes the taker's own offer only on the `DEFAULT` path — `defaultPath_ && offer.quality() >= qualityThreshold_ && owner == strandSrc == strandDst` — so on an autobridged leg the taker's own offer is crossed like any other maker's, its self-transfers netting to nothing (`accountSend` with `sender == receiver` is a no-op).

Engineering changes

- **fix(ledger): a TrustSet toward a missing issuer is tecNO_DST (finding 161)** (f314dca , xrpl-validator) Ledger #106759500, tx D92BD1028719, and six more from the same account that day: rMY7ERFxNN sets a 100000 "RLUSD" line toward rQhWct2f..., an account that does not exist. rippled's TrustSet::preclaim reads the destination root and, with AMM or SingleAssetVault enabled, returns tecNO_DST when it is missing (TrustSet.cpp:213-216); both amendments are live on mainnet, so mainnet takes the fee and writes nothing else.
- **fix(ledger): a payment cannot ripple through an issuer whose NoRipple flag sits on both lines (finding 162)** (039e57e , xrpl-validator) Ledger #106779252, tx 3AD7EA863312, and six more that morning: rJEvC4cuk pays 2,000,000 XRPFLORIDAGATORS to rpFYjv6SF. Both trust lines exist, the sender holds 99,999,999,999 and the destination's limit is ample, yet mainnet returns tecPATH_DRY and charges only the fee. The issuer, rMpjR1oh, carries its NoRipple flag on its side of both holders' lines, and the default path ripples through the issuer: ...
- **fix(ledger): fence the C++ excerpt in check_no_ripple's docs so rustdoc does not compile it** (ddbc941 , xrpl-validator) gate67 failed one test of 659: rustdoc took the indented rippled excerpt in finding 162's doc comment for a Rust doctest. A text fence keeps it prose.
- **fix(ledger): a token escrow snapshots the issuer's transfer rate and finishes net of it (finding 164)** (fd8379b , xrpl-validator) Ledger #106758324, tx 330B52F56E04: r4uNAZYC2k escrows 10 BST under a crypto-condition; BST's issuer charges a 1.08 transfer rate. Ledger #106758384, tx F578BE463712: rpcjmdWFCQ finishes it with the fulfillment.
- **fix(ledger): a live tip beyond the strict limit but inside the inflated one is attempted, not a stop (finding 163)** (20bd643 , xrpl-validator) Ledger #106771950, tx B1283A3C1288, again at #106772087 and #106772093: rBTwLga3i2 buys 1,000,000 XRP with 1,380,000.5 USD.Bitstamp, a limit of 1.3800005e-6 USD per drop. The book's tip is rsRdjxq2's 2208-drop dust offer, FILED at 1.3800013e-6 (a hair beyond the strict limit, inside the 1.0015-inflated one) while its own amounts, 0.0030459516 USD for 2208 drops, price at 1.3795e-6, inside the strict limit. rippled ...
- **fix(ledger): an IOU source is funded from its original balance less its debits, credits deferred (finding 165)** (dd4c94e , xrpl-validator) Ledger #106772946, tx C9E92CF8532F: rLc1HmTpWg's circular partial payment (it pays itself) buys FUZZY through a book that holds its own offers, so one trust line is credited as the destination and debited as the maker inside a single transaction.
- **fix(ledger): a tfSell past its minimum still rests the unsold remainder, judged after repricing (finding 172)** (6283d40 , xrpl-validator) #106758110 E664290A: 7559 BAYN tfSell; the pool took 7456.036 for 124.14 XRP — past the 118.68 minimum — and mainnet rested 102.963884945252 BAYN for 1616532 drops. rippled judges "Offer fully crossed" on the amounts that would rest (OfferCreate.cpp:778), and for tfSell the out side is re-derived from the unsold in first (flowCross :489-506, afterCross.out = divRoundStrict(afterCross.in, rate)). Our guard read the ...
- **fix(ledger): an issuer-owned maker still grosses the taker's spend by the transfer rate (finding 171)** (c1c1980 , xrpl-validator) #106758320 A092BA66 (tfSell|tfIoC selling 18.5185185185186 BST at rate 1.08 into the ISSUER's own offer): the in-side rate is waived only when the offer's owner is the strand source — alice paying alice (BookOfferCrossingStep::getOfInRate) — never because the owner is the issuer. The taker's DirectStep hands the issuer the gross; the issuer → owner send is the no-op. Mainnet debits the taker 20.0000 and the issuer's ...
- **fix(ledger): a book tip beyond the limit does not drop the direct strand when the pool's own bound clears it (finding 169a)** (d00a396 , xrpl-validator) #106766924 961C38C7 (IoC XLM → CNY, iteration 1): activateNext's bound is tip() — the AMM offer when it beats the LOB tip, else the LOB tip (BookStep.cpp:921-953). A LOB tip beyond the limit makes qualityThreshold nullopt, so getOffer emits the maxOffer (quality = the feeless spot) or

the Fibonacci slice, and the strand lives while that bound clears the limit. The next direct offer sat at 0.83737 XLM/CNY against the ...

- **fix(ledger): an LP-token line is torn down only in rippleCreditIOU's full default state (finding 168)** (128b564 , xrpl-validator) #106766906 29526F1B (AMMWithdraw tfLPToken): rEWLR7's LP line carried lsfLowNoRipple while the account has DefaultRipple set — a non-default line — so mainnet zeroed and KEPT it (OwnerCount 709 untouched). We deleted it and took 708. `tear_down_lp_line` checked only the limit and the qualities; it now applies the whole guard (TokenHelpers.cpp:702-716): the LP's reserve bit set, its NoRipple bit differing from the ...
- **fix(ledger): TrustSet creates lines with auth/freeze/deep-freeze bits and enforces the freeze invariants (finding 167)** (48928e9 , xrpl-validator) #106761226 63F07E36 and #106773044 05F705D5: an issuer authorising a holder that has no line yet CREATES it with lsfLowAuth — `trustCreate` carries bAuth, bFreeze and bDeepFreeze onto the creator's side (RippleStateHelpers.cpp:262-277, TrustSet.cpp:735-738). Our creation path only minted the reserve and NoRipple bits: mainnet 0x250000, ours 0x210000, invisible to every per-tx leg but the replay's byte diff.
- **fix(ledger): the AMM limit judge follows limitOut alone and tip()'s quality function (finding 166)** (51a6221 , xrpl-validator) Two crossings the deploy66 soak caught in the single-path AMM tail: a) #106759265 7556BE4A (IoC sell): adjustedRemOut is settled by limitOut ALONE (StrandFlow.h:664-672) before the passes cap the input. We reset `adjusted` on the input clamp, judged the clamped fill strictly and killed a 1.4e-14 miss rippled forgives (tecKILLED vs tesSUCCESS).
- **fix(ledger): a payment's sender is bounded by the deferred-debits fold, and a funds-limited fill clamps to the offer's own TakerPays (findings 169b, 169c)** (059b37e , xrpl-validator) 169b — #106777783 6EF9AF7D and #106773978 D33DB4C8 (tfPartialPayment IOU → XRP sweeps): rippled's per-iteration DirectStep bound is PaymentSandbox::balanceHookIOU, $\min(\text{live}, \text{origBalance} - \Sigma \text{debits})$ with the debits a 16-digit half-even fold. Its trace shows the last DirectStepI::rev bounded at 50251.6343549063 while the line reads 50251.63435490632, and the line keeps 2e-11. The walk's line-bound (held less this level's ...
- **fix(ledger): a forward surplus keeps the reverse cache, a crossing's remainingOut folds to sixteen digits, and the sender bound is the SendMax hop's alone (findings 170, 173, 169b)** (fab8f1e , xrpl-validator) 170 — #106763015 360F1D30 (1e6 drops → GBP through the pool, issuer rate 1.002): DirectStepI::setCacheLimiting keeps the REVERSE cache whenever the forward pass would exceed it — $\text{fwdIn} > \text{cache.in}$, srcToDst or out likewise. rippled's trace: rev asks 1.678428412228995 for 1.67507825571756, the pool's forward swap yields 1.678428412229, and the credit is the cache's 1.67507825571756 (fwd srcToDst 1.675078255717565 ...
- **fix(ledger): an LP-token withdrawal snaps the requested tokens to the pool balance's precision (finding 175)** (188fda9 , xrpl-validator) #106782285 A4B18D75 (AMMWithdraw tfLPToken): fixAMMv1_3's `adjustLPTokens` (AMMHelpers.cpp:181-189, via AMMWithdraw.cpp:752-761, WithdrawAll exempt) rounds the request to what the pool's LPTokenBalance can represent — $(\text{lpTokens} - \text{lptAMMBalance}) + \text{lptAMMBalance}$ under Downward rounding. rPLvYSKR asked 38534539.5663724 of a 1143652631.196906 pool; rippled burns 38534539.566372, the LP keeps 0.0000004 and its line ...
- **fix(ledger): the bridged leg's remainingOut folds too, and a line-bound fill pays the bound's gross verbatim (findings 173, 174)** (ceabbf8 , xrpl-validator) 173, bridged leg — #106781871 825A156A (XAH → RLUSD crossing through the XRP bridge): the crossing's remainingOut is a sixteen-digit IOUAmount on the bridged leg as on the direct walk. 1.84386153980520|36 folds to ...5204, and rippled's last maker gives 0.528031539805204; the exact remainder had us take ...52036 and left the offer 4e-16 apart.

- **fix(ledger): a carry hop takes back no more than the chain parked on the sender's line (finding 177)** (8a23425 , xrpl-validator) #106733664 37F54060 and #106735554 9BCDD090 (rapido, mixed paths): a carry hop spends what the hop chain parked on the sender's line (finding 98) — a line rippled's strand never touches, the carry riding the intermediaries' lines instead (q127's trace: rpxqUyfn in 0.8379734275750757, fee at rvYAfWj5's issuer → maker send, 3.3e-15 left on rpxqUyfn). Re-grossing the net take-back landed one ulp over the park and ...
- **fix(ledger): undoing a park-created line also reverts its creator's OwnerCount (finding 178)** (5f4bf34 , xrpl-validator) #106735554 9BCDD090 (rapido, a mixed path whose first book feeds an account run): the mixed driver measures the carry by parking it on the sender's line and undoes the park before the run. A line the park CREATED carries its creator's reserve bit, sits in both owner directories and charged the creator's OwnerCount (line_adjust's trustCreate path); undo_inflight_lines forgot the line and its directory entries but ...
- **fix(ledger): a bridge leg's offer ladder runs to rippled's 1000-offer limit (finding 179)** (008363c , xrpl-validator) kMaxOffersToConsume is 1000 offers per BookStep per transaction (BookStep.cpp:60, :1085/:1247); cross_bridged built each leg's ladder from the first 128. Surfaced by #106743104 F8084760, a 54-iteration XRPH → XRP → RLUSD crawl whose bundle probe went dry at round 50 — that probe turned out to lack the XRPH/XRP book tip rippled anchors the pool to (the offer is never consumed, so it never reaches the metadata or the ...
- **fix(ledger): a pool slice anchored at a level's own quality shares that level's pass (finding 180)** (dda33ab , xrpl-validator) #106770629 FF3E66E6 (XRP → USDC → RLUSD partial payment): rippled's forEachOffer runs tryAMM on the funded tip's quality and then every offer AT that quality in the same pass — ofrQ is the anchored AMM offer's quality, the tip's — so the pool's 28.0574113498 USDC and rLtCVnoj's 1.538480007399999 reach the RLUSD book as one 29.5958913572 and its maker is debited once: 7879.338142785851 – 29.59331668965259 = ...198. The ...
- **fix(scripts): fetch_tx_bundle seeds explicit-path books and pools, 200-offer book depth** (962e81b , xrpl-validator) Two bundle blind spots that read as engine residues in the s/q sweeps: * EXPLICIT PATHS. Only the direct and XRP-bridge books were seeded, so a Payment whose Paths hop through other currencies had no book at all in the bundle and the probe ran the strand DRY. #106758148 0638A180 (XRP → PKEG → FARM) and #106766924 DA053F84 (USDC → PLX → XLM → XRP) — s332/s367 — are green once every currency hop's book (an account hop ...

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_3_0	2026-08-28 11:15 UTC	2026-09-11 11:15 UTC

Amendments enabled on mainnet: 93.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Deferred Ledger-Triggered Transaction Execution** (payments) Add a new owner-reserved ledger object `ltDEFERRED_TX` (`keylet::deferredTx(account, sequence)`) created by a new `DeferredTxCreate` transaction, storing a
- **Live Credential Re-Check on MPT Transfer** (compliance) Add an `IsfRequireTransferCredential` flag and `sfTransferCredentialIDs` `SField` (`STVector256`, max 8 keylets) to `MPTokenIssuance`, settable only by the issuer
- **Owner Directory Tail Compaction Sweep** (fees) Add a new transaction `ttDIRECTORY_COMPACT` that walks up to 32 entries from the tail (oldest) pages of the invoker's own `OwnerDirectory` and deletes any

Appendix — commits

Time	Repo	Commit	Subject
10:26	xrpl-validator	f314dca	fix(ledger): a TrustSet toward a missing issuer is tecNO_DST (finding 161)
10:32	xrpl-validator	039e57e	fix(ledger): a payment cannot ripple through an issuer whose NoRipple flag sits on both lines (finding 162)
10:53	xrpl-validator	ddbc941	fix(ledger): fence the C++ excerpt in <code>check_no_ripple</code> 's docs so rustdoc does not compile it
11:39	xrpl-validator	fd8379b	fix(ledger): a token escrow snapshots the issuer's transfer rate and finishes net of it (finding 164)
11:39	xrpl-validator	20bd643	fix(ledger): a live tip beyond the strict limit but inside the inflated one is attempted, not a stop (finding 163)
12:27	xrpl-validator	dd4c94e	fix(ledger): an IOU source is funded from its original balance less its debits, credits deferred (finding 165)
13:12	xrpl-validator	6283d40	fix(ledger): a <code>tfSell</code> past its minimum still rests the unsold remainder, judged after repricing (finding 172)
13:12	xrpl-validator	c1c1980	fix(ledger): an issuer-owned maker still grosses the taker's spend by the transfer rate (finding 171)
13:12	xrpl-validator	d00a396	fix(ledger): a book tip beyond the limit does not drop the direct strand when the pool's own bound clears it (finding 169a)
13:12	xrpl-validator	128b564	fix(ledger): an LP-token line is torn down only in <code>rippleCreditIOU</code> 's full default state (finding 168)
13:12	xrpl-validator	48928e9	fix(ledger): <code>TrustSet</code> creates lines with <code>auth/freeze/deep-freeze</code> bits and enforces the freeze invariants (finding 167)
13:12	xrpl-validator	51a6221	fix(ledger): the AMM limit judge follows <code>limitOut</code> alone and <code>tip()</code> 's quality function (finding 166)
13:15	xrpl-validator	059b37e	fix(ledger): a payment's sender is bounded by the deferred-debits fold, and a funds-limited fill clamps to the offer's own <code>TakerPays</code> (findings 169b, 169c)
13:18	xrpl-validator	fab8f1e	fix(ledger): a forward surplus keeps the reverse cache, a crossing's <code>remainingOut</code> folds to sixteen digits, and the sender bound is the <code>SendMax</code> hop's alone (findings 170, 173, 169b)
13:39	xrpl-validator	188fda9	fix(ledger): an LP-token withdrawal snaps the requested tokens to the pool balance's precision (finding 175)
13:39	xrpl-validator	ceabbf8	fix(ledger): the bridged leg's <code>remainingOut</code> folds too, and a line-bound fill pays the bound's gross verbatim (findings 173, 174)
14:04	xrpl-validator	8a23425	fix(ledger): a carry hop takes back no more than the chain parked on the sender's line (finding 177)

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14:13	xrpl-validator	5f4bf34	fix(ledger): undoing a park-created line also reverts its creator's OwnerCount (finding 178)
14:18	xrpl-validator	008363c	fix(ledger): a bridge leg's offer ladder runs to rippled's 1000-offer limit (finding 179)
14:25	xrpl-validator	dda33ab	fix(ledger): a pool slice anchored at a level's own quality shares that level's pass (finding 180)
15:32	xrpl-validator	962e81b	fix(scripts): fetch_tx_bundle seeds explicit-path books and pools, 200-offer book depth
15:32	xrpl-validator	30d4e7a	fix(xrpl-ledger): F181 — a holder-less sender ISSUES to the issuer within its limit
17:53	xrpl-validator	944aeb1	fix(xrpl-ledger): F182 — CredentialCreate needs the reserve for one more object
19:48	xrpl-validator	5449dfa	fix(xrpl-ledger): F183 — a strand's spend is the walk's own figure, not the line's quantised delta
23:14	xrpl-validator	199e7d4	fix(xrpl-ledger): F184 — an autobridged leg crosses the taker's own offer